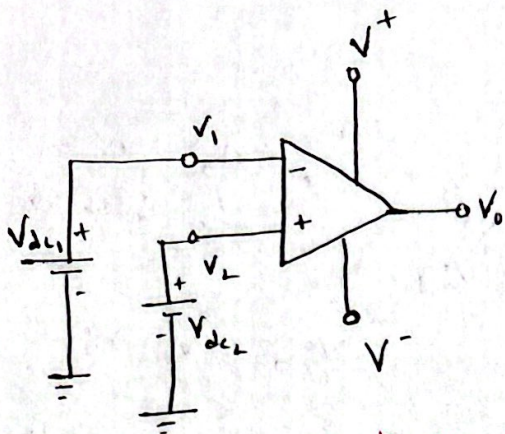
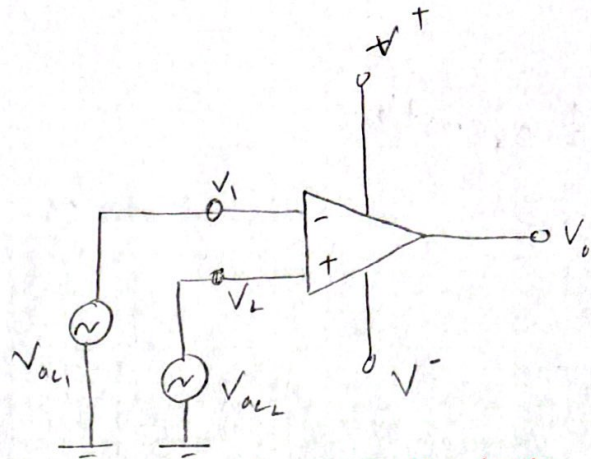


①

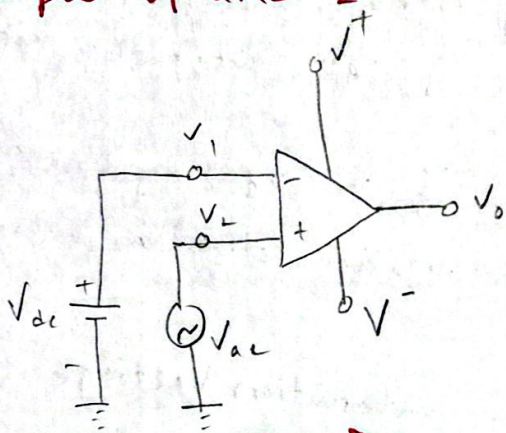
Op-Amp Open-loop Configuration:



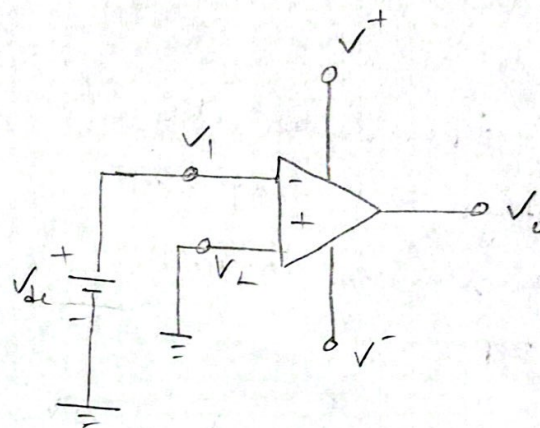
Input V_1 and V_2 both DC



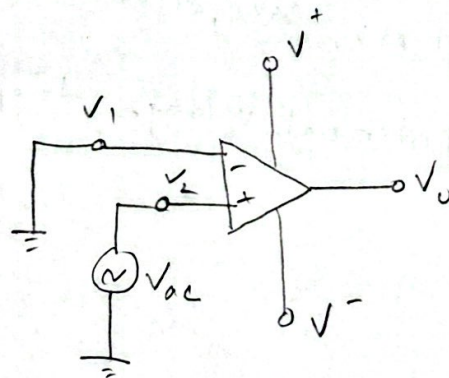
Input V_1 and V_2 both AC



Input V_1 is DC
Input V_2 is AC



Input V_1 is DC
Input $V_2 = 0$



Input $V_1 = 0$
Input V_2 is AC

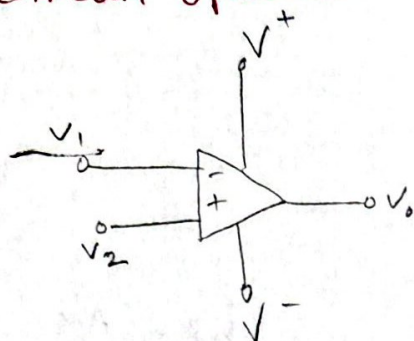
Features:

- ① Output can be only two, either positive or negative
- ② Behaves like a digital circuit, with two output either HIGH or LOW.
- ③ Can't be used for amplifying operation.
 $\Rightarrow$ output doesn't depend on absolute magnitude rather than on their relative magnitude.

Application:

→ switches, comparators, voltage level detector.

Circuit operation:



V_1, V_2 either ac or dc

$$V_o = A_{OL}(V_2 - V_1) \quad , \quad A_{OL} = \infty \text{ for ideal Op-Amp}$$

$$= A_{OL} \& V_{id}$$

$$V_{id} = V_2 - V_1$$

$$= V_{sat}^+ \text{ , if } V_2 > V_1 \text{ , positive } V_{id}$$

↳ positive saturation voltage

$$= V_{sat}^- \text{ , if } V_2 < V_1 \text{ , negative } V_{id}$$

↳ negative saturation voltage

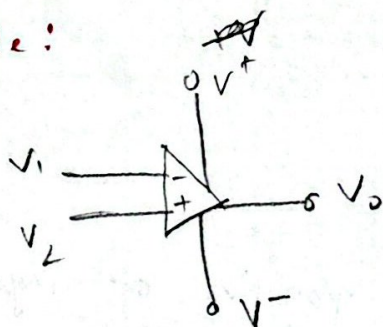
* V_{sat} will depend on the supply voltage but may be some value less for internal voltage drop.

$$V_{sat}^+ = V^+ - \Delta V, \text{ Positive saturation Voltage}$$

$$V_{sat}^- = V^- + \Delta V, \text{ Negative saturation voltage.}$$

ΔV = Internal voltage drop

Example:



Given,

$$V^+ = 12V$$

$$V^- = 12V$$

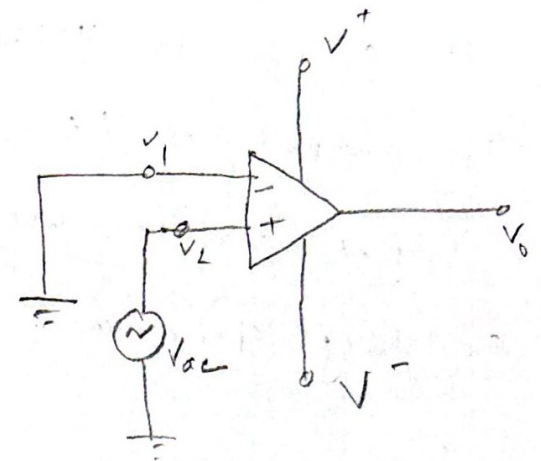
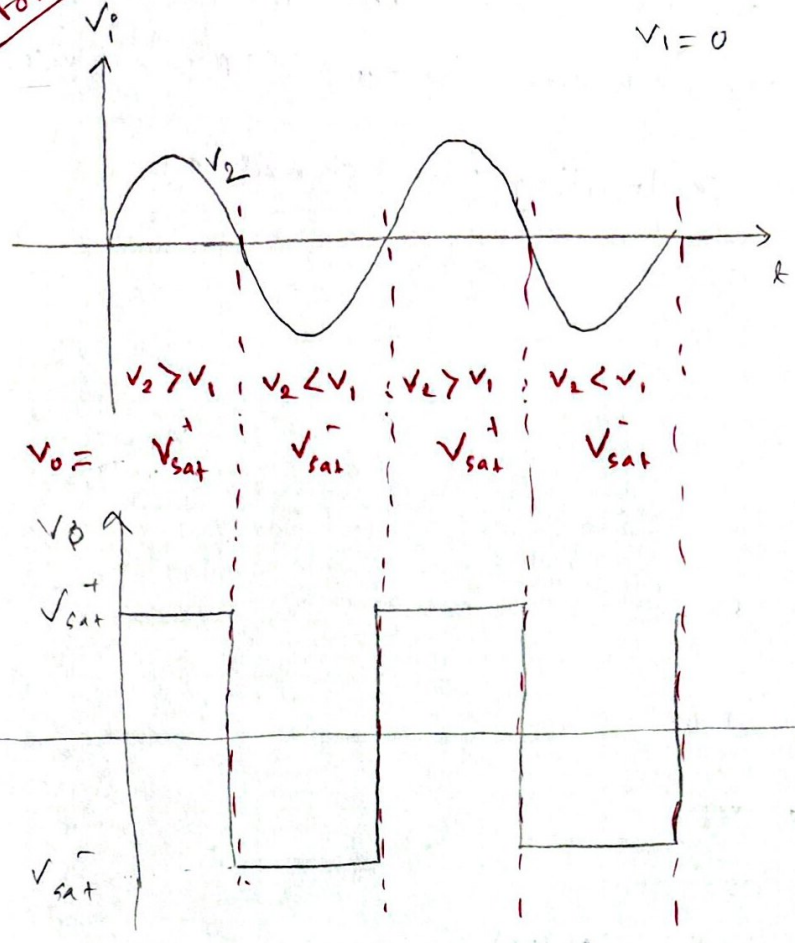
$$\Delta V = 2V$$

	$V_1(V)$	$V_2(V)$	$V_o(V)$
$V_2 > V_1 \rightarrow 1$	0	3	10
$V_2 < V_1 \rightarrow 2$	0	-3	-10
$V_2 < V_1 \rightarrow 3$	3	0	10

$$V_{sat}^+ = V^+ - \Delta V = 12 - 2 = 10V$$

$$V_{sat}^- = V^- + \Delta V = -12 + 2 = -10V$$

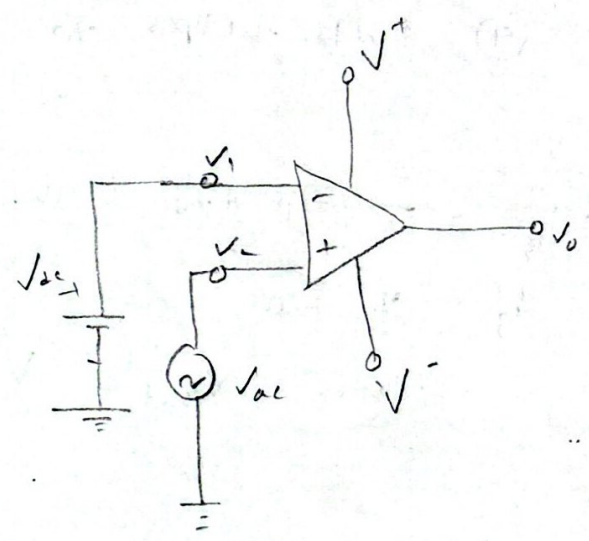
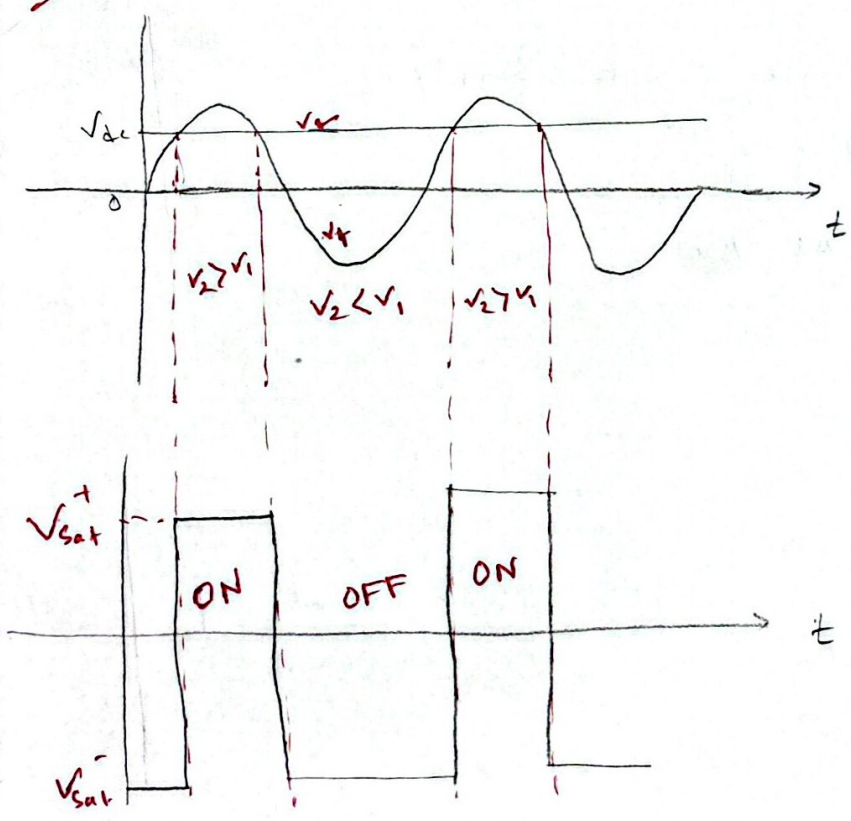
② For AC



$$V_0 = V_{sat}^+, \text{ if } V_2 > V_1$$

$$= V_{sat}^-, \text{ if } V_2 < V_1$$

For AC + DC



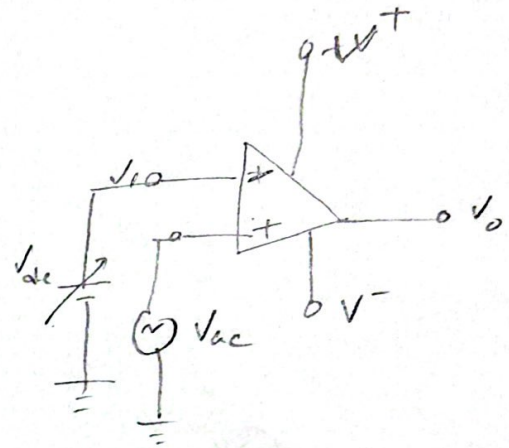
Wave shape same but pulse width, i.e., duty cycle different.

Op-Amp output can be used to turn on and turn off some electrical/electronic equipment.

- # If output is HIGH, it can be used to turn ON the device
- # If output is LOW, it can be used to turn OFF the device.
- # Output can be used for controlling the signal of switching operation.

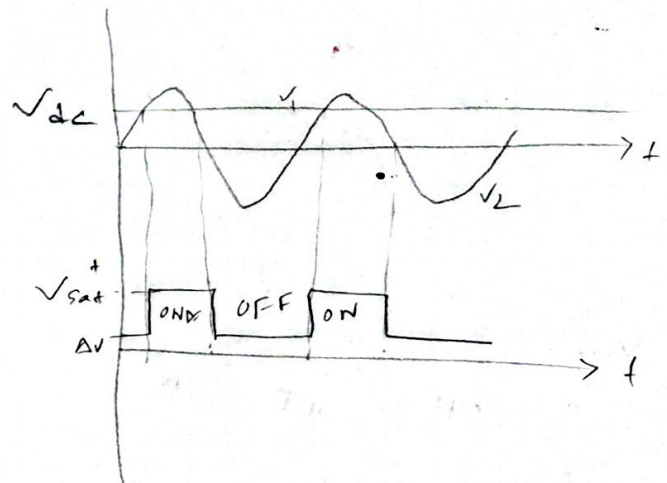
For variable AC and DC

- ① Increasing DC will increase decrease the pulse width (PW)
- ② Decreasing V_{DC} will increase (PW)
- ③ By changing V_{DC} , pulse width can be changed which is known as pulse width modulation (PWM)
- ④ PWM helps to change ON/OFF time.



For $V^- = 0$ → Op-Amp with single power supply.
if $V^- = 0$,

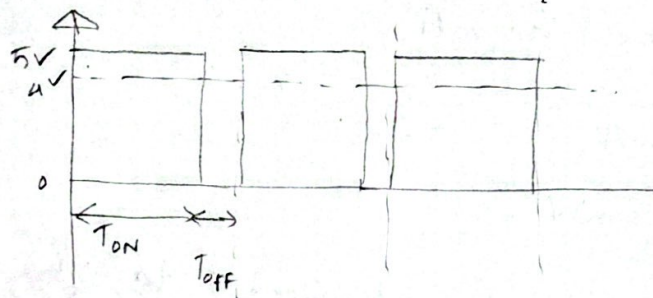
$$\text{then } V_{sat}^- = V^- + \Delta V = \Delta V$$



Application of PWM:

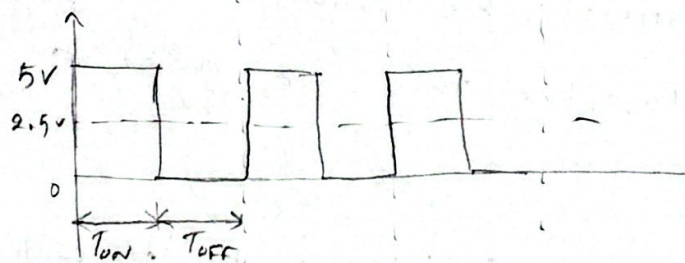
Q. If a lamp set continuous DC supply of 5V, it will glow with a certain intensity but if the DC is applied for 50% of the time and 0V is applied for another 50% of the time its average output voltage will be 2.5V. So, its intensity will be halved.

Ans By modifying the ON/OFF time within a specific period of time, the average value and intensity of the lamp can be varied.



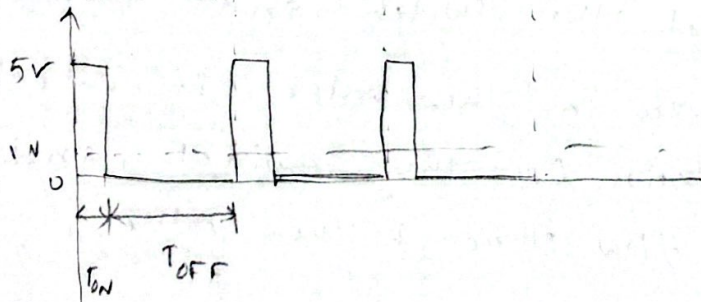
$$\text{Duty Cycle} = \frac{T_{ON}}{T_{ON} + T_{OFF}} = 0.8 (80\%)$$

$$\text{Average voltage} = (5 \times 0.8) \text{ V} = 4 \text{ V}$$



$$\text{Duty Cycle} = 0.5 (50\%)$$

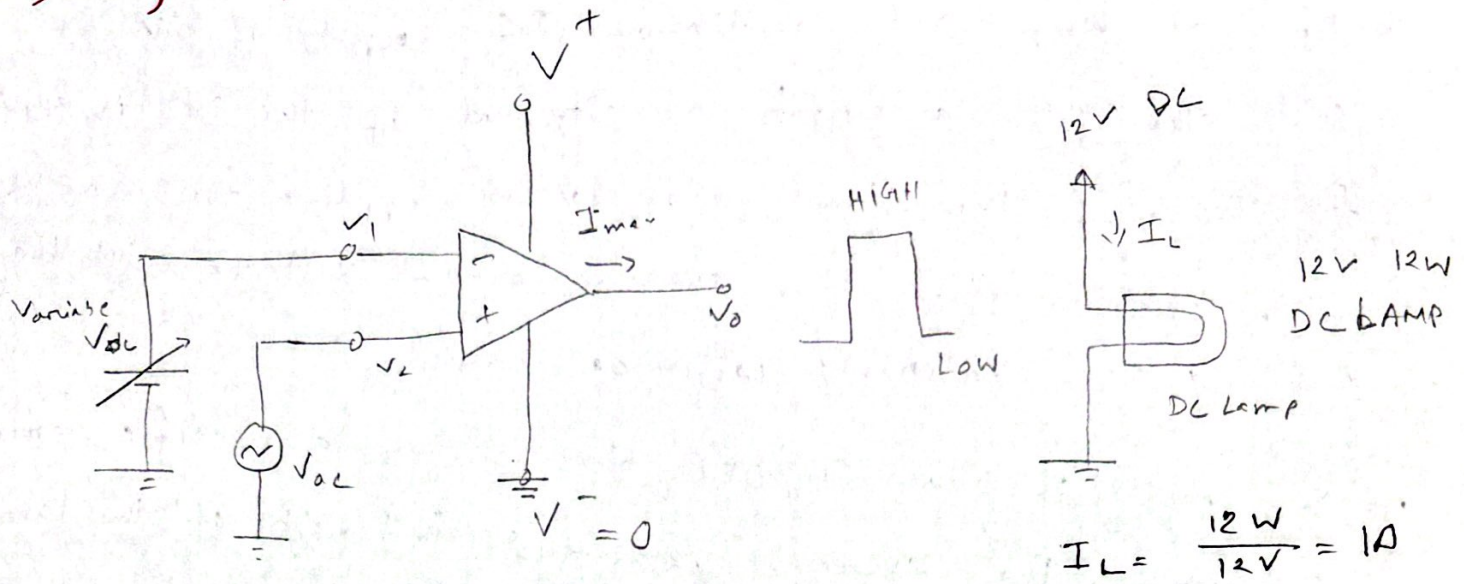
$$\text{Average voltage} = (5 \times 0.5) \text{ V} = 2.5 \text{ V}$$



$$\text{Duty Cycle} = 0.2 (20\%)$$

$$= (5 \times 0.2) \text{ V} = 1 \text{ V}$$

Design of a dimmer circuit



Op-Amp can't handle large power, it can handle power mill amp range.

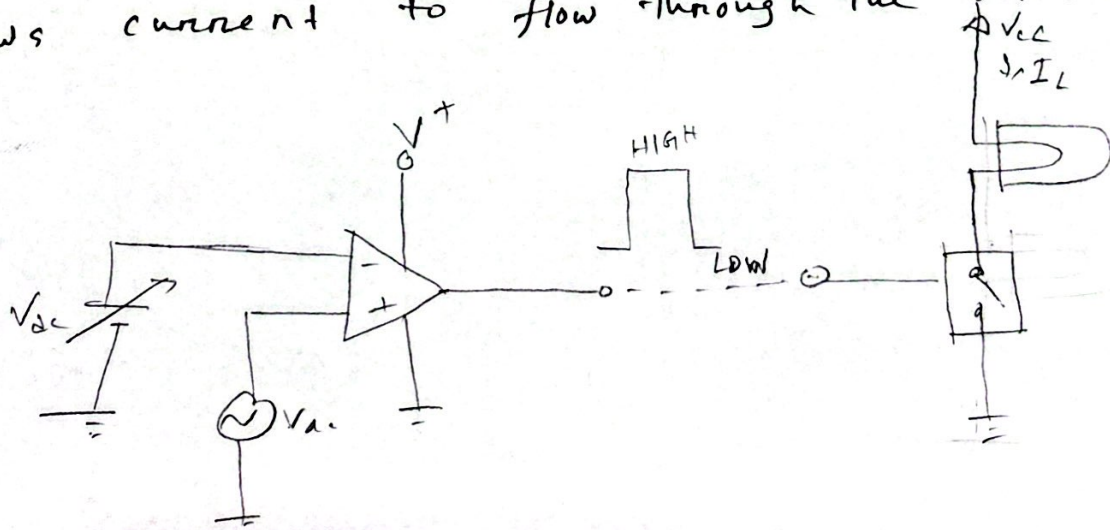
I_{max} can be 20 mA.

$$I_{max} < 20 \text{ mA}$$

P_{max} in mW

So, op-Amp can't directly be connected to the valve to light it up. To solve this we need a switching element.

The Op-Amp will control the switch and it will control the flow of power in the valve. If output of Op-Amp is HIGH, it will turn ON the switch and allows current to flow through the valve.



Two kind of which

① transistor

② electro mechanical relay.

Transistor as a switch:

① Am. Active

OFF → ② Cut off

ON → ③ Saturation

EBJ

FB

RB

FB

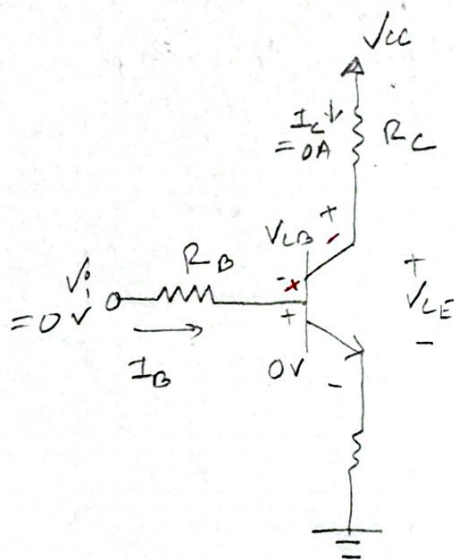
CBJ

RB

RB

FB

} Switching Application



if $V_i = 0V$

Cut off mode

① if, $V_i = 0$,

$V_B = 0$, hence

$I_C = 0$.

Transistor in OFF state (cut off mode)

② if, $V_i = 0$,

$$I_B = \frac{V_i - V_{BE}}{R_B}$$

$$I_C = \beta I_B \Rightarrow I_B = \frac{I_C}{\beta}$$

$$V_{CE} = V_{CC} - I_C R_C$$

$$= V_C > V_B$$

transistor active mode
transistor ON

③ $V_{BE} \approx 0.7V$

But, $V_{BC} \approx 0.4V$

$$V_C < V_B$$

$$V_{CE} = V_{BE} - V_{BC} \approx 0.2V$$

↓
 $V_{CE sat}$

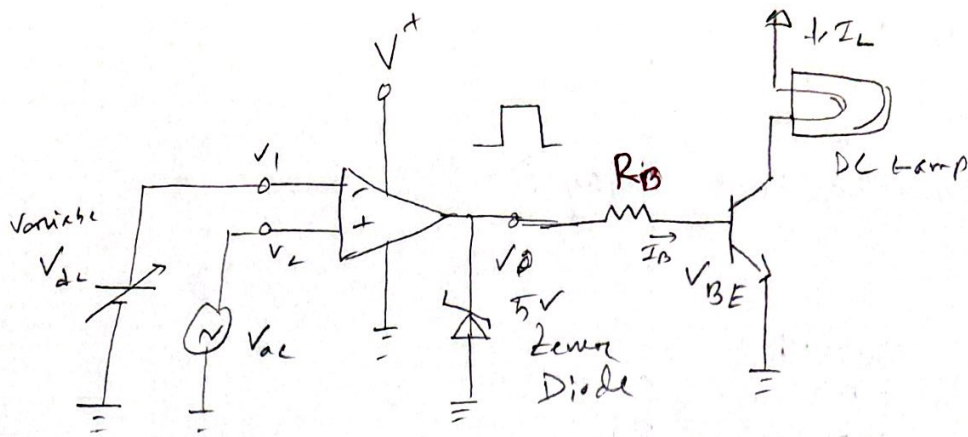
transistor saturation mode.

$$I_B \neq \frac{I_C}{\beta}$$

$$I_B > \frac{I_C}{\beta}$$

$$\Rightarrow I_B = \frac{V_i - V_{BE}}{R_B} > \frac{I_C}{\beta}$$

$$\therefore R_B < \frac{V_i - V_{BE}}{(I_C/\beta)}$$



• Use of power transistor to handle a large power.

A 5V Zener diode is connected to fix the voltage of the output & every time the voltage goes high:

$$I_B > I_C / \beta = I_L / \beta$$

$$R_B < \frac{V_1 - V_{BE}}{(I_L / \beta)}$$

$I_L \rightarrow$ Rated current of lamp.

Maths from Setra/Smith

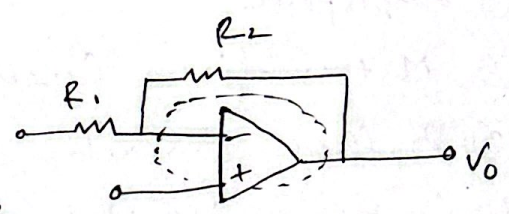
Chapter-6

for choosing values of circuit elements in

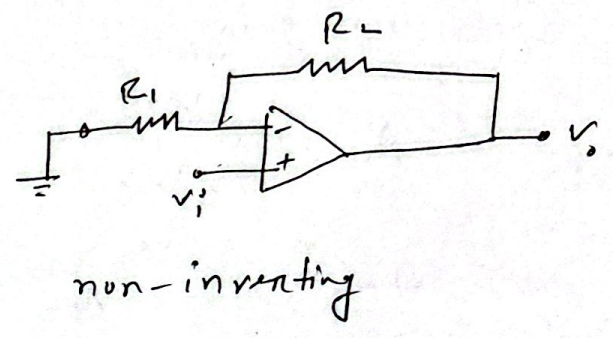
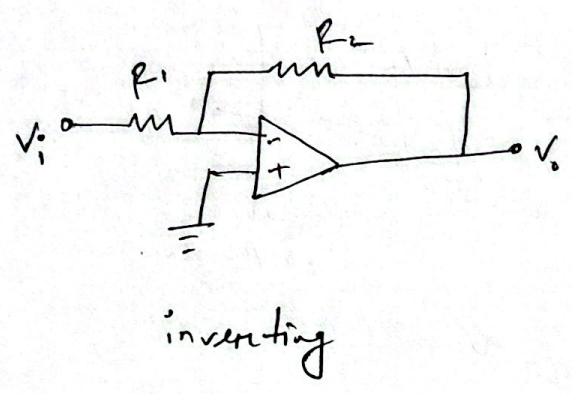
BJT for different mode of operation.

Op-Amp - Close loop Connection!

=> R_L closes the loop around the Op-Amp creating feedback.



=> if input is applied on negative terminal it is inverting amplifier and if input is connected in positive terminal it is non-inverting amplifier.

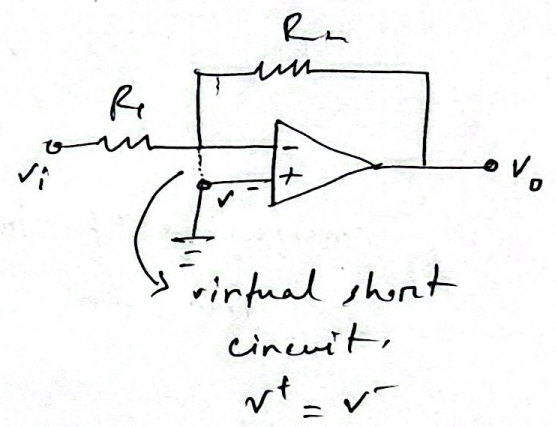


Features:

① Virtual Short Circuit.

Voltage at two terminal in close loop will be same, which is known as virtual short circuit.

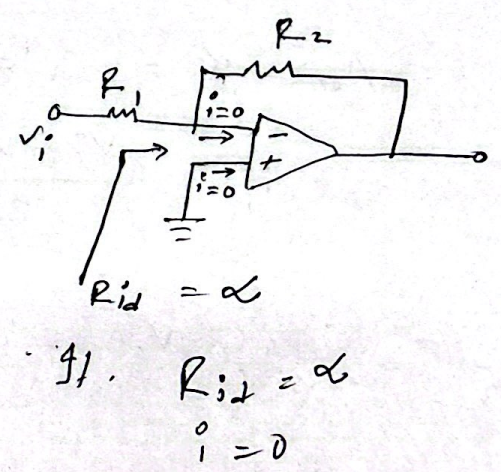
Valid for only close loop connection.



② Zero input current

=> For ideal case i flowing at both terminal of op-amp is zero which implies infinite differential input resistance.

valid for both open and close loop



Application :

- ① Amplifier
- ② Mathematical Circuit
- ③ Buffer circuit
- ④ Filter circuit
- ⑤ Oscillator circuit.

Open loop gain :

$$A_{OL} = \frac{V_o}{V^+ - V^-}$$

$$\Rightarrow V^+ - V^- = \frac{V_o}{A_{OL}} = 0$$

$\Downarrow$
for ideal case $A_{OL} = \infty$

$$\therefore V^+ - V^- = 0$$

$$\Rightarrow V^+ = V^-$$

(virtual short circuit)

consequence of infinite loop gain A_{OL}

True for close loop connection with ideal op-amp.

Output voltage and gain:

$$R_{id} = \infty \Rightarrow i = 0$$

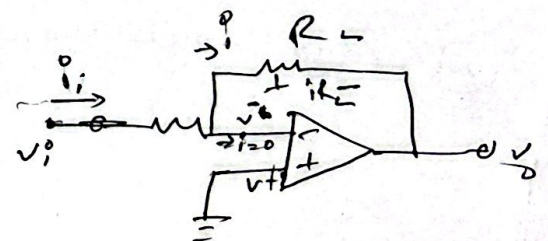
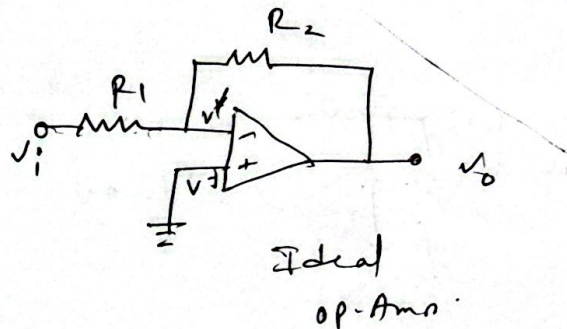
$$A_{OL} = \infty \Rightarrow V^+ = V^-$$

$$i_i = \frac{V_i - V^-}{R_1} = \frac{V_i}{R_1}$$

$$[\because V^- = V^+ = 0]$$

$$\therefore V_o = -i R_2 + V^-$$

$$= -i R_2 \quad (i = i_i, V^- = 0)$$



$$\Rightarrow V_o = -\frac{V_i}{R_1} R_2 \quad \left(i_i = \frac{V_i}{R_1} \right)$$

$$\therefore \text{close loop gain } A_{cl} = \frac{V_o}{V_i} = -\frac{R_2}{R_1}$$

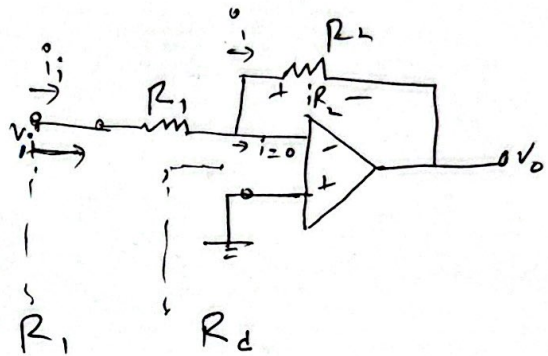
As gain is $(-)$, inverting amplifier.

Input Resistance:

R_d = differential input resistance

R_1 = Input resistance of the amplifier seen from the source.

$$R_i = \frac{V_i}{i_i} = R_1 \quad \left[i_i = \frac{V_i}{R_1} \right]$$



→ Finite input resistance R_1 for Inverting Amplifier.

R_1 is considered to be low and it will

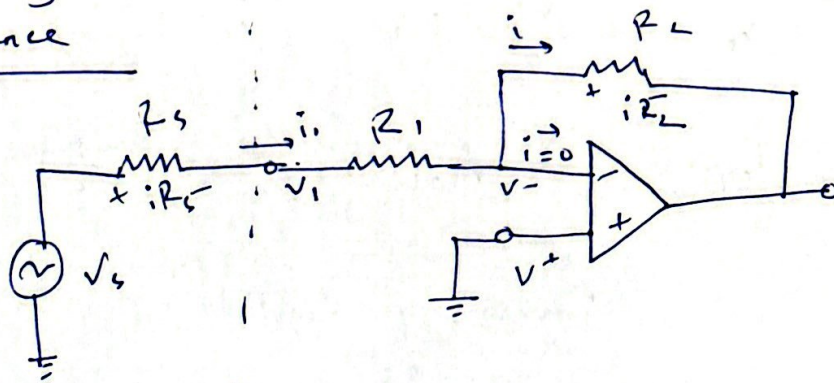
limit the gain.

$$A_{cl} = \frac{V_o}{V_i} = -\frac{R_2}{R_1}$$

≠ R_1 should be low and R_2 has to be high to maintain high gain.

Practical Signal Source

Inverting Amp



$$i_i = \frac{V_i}{R_i} = \frac{V_i}{R_1}, \quad R_i = R_1$$

$$V_i = V_s - i_i R_s < V_s$$

$$R_1 \downarrow, i_i \uparrow, i_i R_s \uparrow, V_i \downarrow, V_o \downarrow \quad (V_o = A_{cl} V_i)$$

High i_i , High $i_i^2 R$ loss

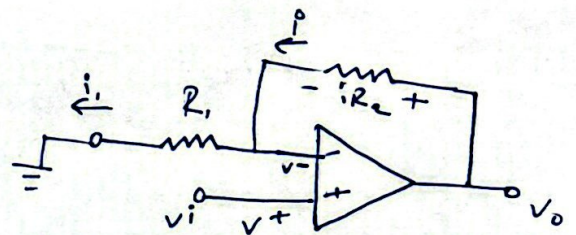
Loss of signal at the input.

Low input resistance R_i degrades the performance and reduces the lifetime of the device.

Non-Inverting Amp:

$$R_{id} = \infty, \quad i = 0$$

$$A_{OL} = \infty, \quad V^+ = V^-$$



$$i_i = \frac{V^-}{R_1} = \frac{V_i}{R_1} \quad (V^- = V^+ = V_i)$$

$$V_o = i R_2 + V^- \quad (i = i_i = \frac{V_i}{R_1})$$

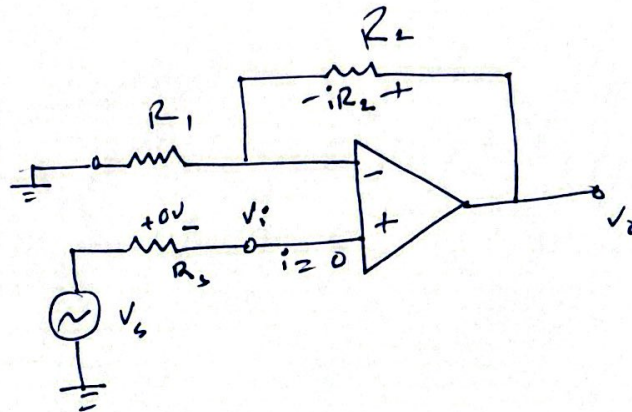
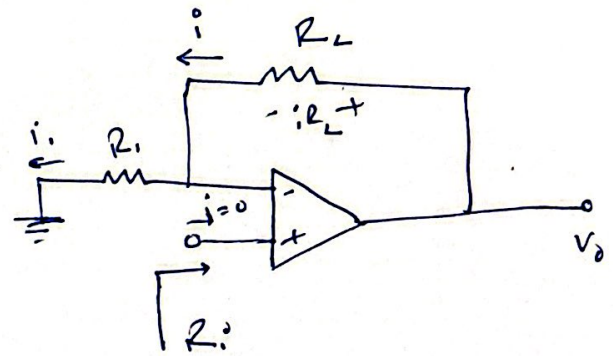
$$= \frac{V_i}{R_1} R_2 + V_i$$

$$= V_i \left(1 + \frac{R_2}{R_1} \right)$$

$$\therefore A_{cl} = \frac{V_o}{V_i} = 1 + \frac{R_2}{R_1}$$

$$R_i = \frac{V_i}{i_i} = \frac{V_i}{i} = \frac{V_i}{0} = \infty$$

∴ Non-inverting Amp has ideally infinite input resistance.



$$R_i = \infty, \quad i = 0 \Rightarrow V_i = V_s$$

$$\neq \quad i = 0, \text{ No. } i^2 R \text{ loss}$$

$$\neq \quad \text{No loss of signal at the input.}$$